

Introduction To 10s Complement

The Logic

Digital hardware struggles with standard subtraction due to complex borrowing logic.

Solution: The 10s complement method converts subtraction entirely into addition.

The Rules

Goal: Compute $A - B$

Rule: Add A to the 10s complement of B .

Formula: 10s complement of $B = 10 - B$

Case 1: Positive Result

$$7 - 4$$

Step 1: Find 10s complement of 4.

$$10 - 4 = 6$$

Step 2: Add result to 7.

$$7 + 6 = 13$$

Step 3: Analyse carry digit.

Sum is 13. Carry is 1.

Step 4: Discard carry 1.

Final answer = 3

Case 2: Negative Result

$$4 - 7$$

Step 1: Find 10s complement of 7.

$$10 - 7 = 3$$

Step 2: Add result to 4.

$$4 + 3 = 7$$

Step 3: Analyse carry digit.

Sum is 7. Carry is 0.

Step 4: 0 carry means negative result.
Take 10s complement of 7.

$$10 - 7 = 3$$

Step 5: Append minus sign.

$$\text{Final answer} = -3$$

Hardware Requirements

10s Complement Generator

Role: Computes **10** - B.

Hardware: Built using **1** **4**-bit binary adder and XOR logic gates.

BCD Adder

Role: Adds A to the generated **10**s complement of B.

Hardware: Built using **2** **4**-bit adders. Internal logic adds **6** when initial sum **> 9**.